

Exercise 21A — Full Solutions

Questions 1–8 (complete)

Implicit differentiation: first and second derivatives

Colour key: *green = verification check* | *blue = alternative method*. Throughout, y' denotes $\frac{dy}{dx}$ and y'' denotes $\frac{d^2y}{dx^2}$.

Question 1

Find $\frac{dy}{dx}$ in each case.

(a) $xy = e^x$. Product rule on the left:

$$xy' + y = e^x \implies \boxed{\frac{dy}{dx} = \frac{e^x - y}{x}}.$$

(b) $y^2x = e^y$. Product rule (left), chain rule (right):

$$2xy y' + y^2 = e^y y' \implies y'(2xy - e^y) = -y^2 \implies \boxed{\frac{dy}{dx} = \frac{y^2}{e^y - 2xy}}.$$

(c) $\tan(x + y) = y$. Chain rule on the left:

$$\sec^2(x + y)(1 + y') = y' \implies \sec^2(x + y) = y'(1 - \sec^2(x + y)).$$

Since $1 - \sec^2 \theta = -\tan^2 \theta$,

$$\boxed{\frac{dy}{dx} = -\frac{\sec^2(x + y)}{\tan^2(x + y)} = -\operatorname{cosec}^2(x + y)}.$$

Alternative — in terms of y . Since $\tan(x + y) = y$, we have $\sec^2(x + y) = 1 + y^2$ and $\tan^2(x + y) = y^2$, so $\frac{dy}{dx} = -\frac{1 + y^2}{y^2}$.

Question 2

Find $\frac{d^2y}{dx^2}$ of $x^2 = ye^x + y^2$ at $(0, 0)$.

Check the point lies on the curve. At $(0, 0)$: LHS = $0^2 = 0$; RHS = $0 \cdot e^0 + 0^2 = 0$. ✓

First derivative (product rule on ye^x , chain rule on y^2):

$$2x = e^x y' + ye^x + 2yy'.$$

At $(0, 0)$: $0 = (1)y' \Rightarrow y' = 0$. Differentiate again:

$$2 = e^x(y' + y'') + e^x(y' + y) + 2((y')^2 + yy'').$$

At $(0, 0)$ with $y' = 0$: $2 = y''$, so

$$\boxed{\frac{d^2y}{dx^2} = 2}.$$

Question 3

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ of $\ln(x+y) = 2y$ at $(1, 0)$.

Check the point lies on the curve. At $(1, 0)$: LHS = $\ln(1+0) = 0$; RHS = $2(0) = 0$. ✓

First derivative:

$$\frac{1+y'}{x+y} = 2y' \implies 1+y' = 2y'(x+y).$$

At $(1, 0)$: $1+y' = 2y' \Rightarrow y' = 1$. Differentiate $1+y' = 2y'(x+y)$:

$$y'' = 2y''(x+y) + 2y'(1+y').$$

At $(1, 0)$ with $y' = 1$: $y'' = 2y'' + 4 \Rightarrow y'' = -4$. Hence

$$\boxed{\frac{dy}{dx} = 1}, \quad \boxed{\frac{d^2y}{dx^2} = -4}.$$

Question 4

Given $xy = \sin(x+y)$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

First derivative:

$$xy' + y = \cos(x+y)(1+y') \implies \boxed{\frac{dy}{dx} = \frac{\cos(x+y) - y}{x - \cos(x+y)}}.$$

Differentiate $xy' + y = \cos(x+y)(1+y')$ again:

$$xy'' + 2y' = -\sin(x+y)(1+y')^2 + \cos(x+y)y''.$$

Solving for y'' :

$$\boxed{\frac{d^2y}{dx^2} = -\frac{\sin(x+y)(1+y')^2 + 2y'}{x - \cos(x+y)}} \quad (y' \text{ as above}).$$

Question 5

The curve $C : y^2 = \ln x + e^y - 1$ passes through $(1, 0)$. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ there.

Check the point lies on the curve. At $(1, 0)$: LHS = 0 ; RHS = $\ln 1 + e^0 - 1 = 0 + 1 - 1 = 0$. ✓

First derivative:

$$2yy' = \frac{1}{x} + e^y y'.$$

At $(1, 0)$: $0 = 1 + y' \Rightarrow y' = -1$. Differentiate again:

$$2(y')^2 + 2yy'' = -\frac{1}{x^2} + e^y(y')^2 + e^y y''.$$

At $(1, 0)$ with $y' = -1$: $2 = -1 + 1 + y'' \Rightarrow y'' = 2$. Hence

$$\boxed{\frac{dy}{dx} = -1}, \quad \boxed{\frac{d^2y}{dx^2} = 2}.$$

Question 6

A curve is given by $xy^\alpha = \beta$ (α, β constants).

(a) Differentiate (right side constant):

$$y^\alpha + \alpha xy^{\alpha-1}y' = 0 \implies \boxed{\frac{dy}{dx} = -\frac{y}{\alpha x}}.$$

(b) Differentiate $y' = -\frac{1}{\alpha} \cdot \frac{y}{x}$ (quotient rule):

$$y'' = -\frac{1}{\alpha} \cdot \frac{xy' - y}{x^2}.$$

Substitute $xy' = -\frac{y}{\alpha}$, so $xy' - y = -y\frac{1+\alpha}{\alpha}$:

$$y'' = -\frac{1}{\alpha} \cdot \frac{-y(1+\alpha)/\alpha}{x^2} = \boxed{\frac{y(1+\alpha)}{\alpha^2 x^2}}. \quad \blacksquare$$

Check (take $\alpha = 1$). $xy = \beta \Rightarrow y = \beta x^{-1}$, giving $\frac{d^2y}{dx^2} = 2\beta x^{-3} = \frac{2y}{x^2}$, which the formula reproduces. ✓

Question 7

The curve $(x+y)^6 = x$ passes through $(1,0)$. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ there.

Check the point lies on the curve. At $(1,0)$: $(1+0)^6 = 1 = x$. ✓

First derivative:

$$6(x+y)^5(1+y') = 1.$$

At $(1,0)$: $6(1+y') = 1 \Rightarrow y' = -\frac{5}{6}$. Differentiate again:

$$30(x+y)^4(1+y')^2 + 6(x+y)^5y'' = 0.$$

At $(1,0)$ with $1+y' = \frac{1}{6}$: $30 \cdot \frac{1}{36} + 6y'' = 0 \Rightarrow \frac{5}{6} + 6y'' = 0 \Rightarrow y'' = -\frac{5}{36}$. Hence

$$\boxed{\frac{dy}{dx} = -\frac{5}{6}}, \quad \boxed{\frac{d^2y}{dx^2} = -\frac{5}{36}}.$$

Question 8

The curve $\sin 2x \cos 2y = \frac{\sqrt{3}}{4}$ passes through $P\left(\frac{\pi}{3}, \frac{\pi}{6}\right)$. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at P .

Check the point lies on the curve. At P : $\sin \frac{2\pi}{3} \cos \frac{\pi}{3} = \frac{\sqrt{3}}{2} \cdot \frac{1}{2} = \frac{\sqrt{3}}{4}$. ✓

At P : $\sin 2x = \frac{\sqrt{3}}{2}$, $\cos 2x = -\frac{1}{2}$, $\sin 2y = \frac{\sqrt{3}}{2}$, $\cos 2y = \frac{1}{2}$.

(a) Differentiate, then divide by 2:

$$\cos 2x \cos 2y - \sin 2x \sin 2y y' = 0 \implies y' = \frac{\cos 2x \cos 2y}{\sin 2x \sin 2y}.$$

At P : $y' = \frac{\left(-\frac{1}{2}\right)\left(\frac{1}{2}\right)}{\left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)} = \frac{-1/4}{3/4}$, so

$$\boxed{\frac{dy}{dx} = -\frac{1}{3}}.$$

(b) Differentiate $\cos 2x \cos 2y - \sin 2x \sin 2y y' = 0$:

$$-2 \sin 2x \cos 2y - 4 \cos 2x \sin 2y y' - 2 \sin 2x \cos 2y (y')^2 - \sin 2x \sin 2y y'' = 0.$$

At P with $y' = -\frac{1}{3}$:

$$-\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{3} - \frac{\sqrt{3}}{18} - \frac{3}{4}y'' = 0 \implies y'' = -\frac{8\sqrt{3}}{9} \cdot \frac{4}{3} = \boxed{-\frac{32\sqrt{3}}{27}}.$$